

Sullivan Gleason

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EDUCATION

UNIVERSITY – Southeast Missouri State University C/O 2026, GPA: 4.0

- Computer Science Major – Bachelor of Science
- Mathematics Minor
- Cybersecurity Minor
- Data Science Minor

WORK EXPERIENCE

BOEING — Berkeley, MO

Product Security Engineer

June 2025 – Present

Internal Research and Development for the Boeing Company in the field of product security. Strong emphasis on cryptographic techniques on embedded systems.

GUILFORD TECHNOLOGIES — O’Fallon, MO

Software Developer

August 2023 – November 2023

Developed a cloud-based cybersecurity network scanning tool for businesses and enterprises to secure their networks by the use of automating active and passive *Network Mapper (Nmap)* scans. Scans determine the attack surface and routinely test network edge devices for vulnerabilities as scheduled by the user.

April 2024 – December 2024

Lead front-end developer for a network drawing tool aid to give more flexibility in the planning, deployment, and maintenance of real-world network architectures. Software gives users control of both the breadth and depth of network design, allowing them to build a network from scratch on a dynamic canvas.

Software Development Summer Intern

May 2023 -August 2023

Aided the implementation of a more efficient network architecture for the University of Massachusetts Chan Medical School. Led the development of *NetBox* plugins in Python’s *Django* Framework to automate the scheduling of network structure changes. Gained extensive experience in the field of Computer Networking, Single Source of Truth (SSOT), network configuration management software *NetBox* and Ansible.

ATHLETIC TUTOR — Cape Girardeau, MO

Math, Physics and Computer Science Tutor

August 2023 – December 2024

Tutor fellow student athletes on topics ranging from math fundamentals to advanced calculus and linear algebra. Cover computer science principles such as Object-Oriented Programming (OOP), data types, and algorithms to help build a foundation for their current and future classes. Build an understanding for more advanced topics such as web design, advanced algorithms and optimization, and database management. Open tutoring hours allow several students to be tutored at once, creating a classroom-like atmosphere.

UNDERGRADUATE RESEARCHER — Northridge, CA

Systems Researcher - NSF REU Program

June 2025 – August 2025

Conducted research under the advisement of Dr. Marjan Asadinia at California State University, Northridge. Research focused on binary encoding methods to minimize energy consumption and improve reliability in the experimental Phase Change Memory. Developed robust data transformation pipelines using machine learning algorithms. Leveraged low-overhead data representation using Vector-Quantized Autoencoders. First author of two research papers.

UNDERGRADUATE RESEARCHER — Cape Girardeau, MO

Deep Learning/AI Security

September 2025 – May 2026

Research on generation of real-time adversarial perturbations for computer vision models. Further research on training methods to preserve invertibility in lossy networks. Funded through an NSF research grant under Dr. Ziping Liu at Southeast Missouri State University

PUBLICATIONS

1. **S. Gleason**, P. Kakadiya, M. Asadinia, “Learn2Mask: A Predictive Encoding Framework for Energy-Efficient PCM Writes”. Proceedings of the AIBThings Conference, 2025.
 - a. Published in IEEE Xplore, Presented Online at Central Michigan University
2. **S. Gleason**, P. Kakadiya, M. Asadinia, “Energy-Efficient Encoding for Multi-Level Cell PCM Using VQ-VAE-Based Masking”, Proceedings of the IEEE IEMCON Conference, 2025.
 - a. Pending Publication, Presented at University of California, Berkeley

PERSONAL PROJECTS

NEURAL NETWORK FRAMEWORK (C++ (Eigen))

Implemented, from scratch, a modular neural network framework in C++ with several layer types, update rules, activation functions and loss functions. Custom vector, matrix and N-Dimensional tensor classes. Implemented a Conditional Generative Adversarial Network (cGAN) as proof of functionality.

3D RENDERING ENGINE (Rust, (Vulkano))

Created a 3D rendering engine in Rust from scratch. Consists of an efficient matrix-transformation pipeline for efficient 3D-2D projection, complete with spatial rotations and translations. Harnesses custom GLSL GPU kernels to efficiently render .obj files to the screen concurrently at ~300 fps on a small laptop. Optimized for performance and concurrency on limited hardware.

HARDWARE-BASED HUMAN VERIFICATION WEBCAM

Developed a live video camera to protect sensitive video calls from tampering methods such as deepfakes and replay attacks. Uses a novel combination of depth and video sensors to estimate the probability of such tampering using a learned algorithm. Provisional Patent Application Submitted to US Patent and Trademark Office.

SOFTWARE AND TOOLS

Languages: Python, JavaScript, Java, TypeScript, Rust, Dart, PHP, SQL, C#, C, C++, GLSL

Libraries/Frameworks: Django, React, React Native, PyTorch, Tensorflow, Flutter, Eigen, OpenCV, Vulkano

Hardware: RaspberryPi, Esp32, Arduino

OTHER EXPERIENCE

- *Funeral home pianist*
- *Concession stand cashier*

UNIVERSITY

Athletics and Clubs

- Pole Vaulter | Track and Field team
- Competitive Programming Club
- Preserving Planet (POP) Club Member
- Piano Instructor
- Student Research Committee Planning Board
- University Jazz Band

Awards

- 2024-2025 Outstanding Student in Computer Science
- Ohio Valley Conference Scholar Athlete (One of Six in Conference)
- 2x Ohio Valley Conference Pole Vault Champion, Conference Meet Record Holder
- Southeast Missouri State University Pole Vault Indoor/Outdoor School Record Holder
- NSF CISE REU Travel Grant to attend the 2025 IEEE 16th Annual IEMCON
- Best Presentation – Artificial Intelligence Track - IEEE IEMCON
- NCAA DI All American Honorable Mention

HIGH SCHOOL – Fort Zumwalt North High School C/O 2022

Academics and Awards

- 35 ACT / 1520 SAT
- Mathematics Department Award, German Department Award, German Seal of Biliteracy
- Award of Excellence for Exemplary Display of Character Ethics and Sportsmanship
- 2021, 2022 Class 5 Pole Vault Missouri State Champion, School Record Holder